

# AKSHAYA THOUTAM

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Languages/Tools: Python, R, Bash, JavaScript, HTML, CSS, Jupyter, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Azure, Linux/Unix, GitHub, Shiny

## EDUCATION

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### HARVARD MEDICAL SCHOOL M.S Clinical Research - 4.0 GPA

Cambridge, Ma 2025-2026

- Relevant Coursework: CRES 712: Survey of Modern omics, BETH 720: Ethics in genomics, CI 766: Translation innovation into practice, CI 700: Ethics and the IRB, CI 701: Clinical Data Science Design and Analytics I, CI 702A: Clinical Trials (Part 1), CRES 703: Developing Therapies for Patients, CRES 705: Systematic Review and Meta-Analyses, CRES 709+CRES710: Interpreting Medical Literature (Part 1 and 2)

### GEORGIA INSTITUTE OF TECHNOLOGY B.S Neuroscience

Atlanta, Ga 2020-2023

- Minor in Health and Medical Sciences
- Relevant Coursework: Linear Algebra, Integral Calculus, Biostatistics, Machine Learning in Bioscience, Cancer Biology and Technology, Tumor Microenvironment and Immuno-Oncology (MIT), Advanced Computational Biology: Genomes Networks Evolution (MIT), Statistical Inference for Brain and Cognitive Sciences (MIT), Artificial Intelligence in Medicine II (Harvard)

## PUBLICATIONS First Author †

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1. Alan DenAdel, Madeline Hughes, **Akshaya Thoutam**, Ava P. Amini, and Lorin Crawford. Evaluating the role of pre-training dataset size and diversity on single-cell foundation model performance. (2026). *Nature Methods*.
2. Zeinab Navidi, **Akshaya Thoutam**, Madeline Hughes, Srivatsan Raghavan, Peter Winter, Lorin Crawford, and Ava P. Amini. Adaptive Resampling For Improved Cellular Representation Learning in Imbalanced Single-cell Samples. (2025). *Peer review submitted*.
3. Ajay Nadig & **Akshaya Thoutam†**, Madeline Hughes, Srivatsan Raghavan, Peter Winter, Ava P. Amini, and Lorin Crawford. Consequences of training data composition for deep generative models in single cell biology (2025). *Peer Review submitted*.
4. Greg Gibson, Dermot McGovern, **Akshaya Thoutam**, the NIDDK IBD Genetics Consortium, Judy Cho and John Rioux. Eleven Grand Challenges for Inflammatory Bowel Disease Genetics and Genomics. (2024). *Inflammatory Bowel Disease*
5. Nasrin Hooshmand, **Akshaya Thoutam†**, Max Anikovskiy, Hagar Labouta C Mostafa El-Sayed (2021). Localized Surface Plasmon Resonance as a Tool to Study Protein Corona Formation on Nanoparticles. *Journal of Physical Chemistry C*, 125(45), 24765–24776
6. **Akshaya Thoutam†**, Narasiah C Kolliputi (2021). Epigenetics of pulmonary diseases. In T. Tollefsbol (Ed.), *Medical Epigenetics* (2nd ed., pp. 45-67). Publisher. ISBN: 9780128239285
7. **Akshaya Thoutam†**, Mason Breitzig, Richard Lockey C Narasiah Kolliputi (2020). Coronavirus: a shift in focus away from IFN response and towards other inflammatory targets. *Journal of cell communication and signaling*

## RESEARCH EXPERIENCE

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### Project Ex Vivo | **BROAD INSTITUTE OF HARVARD/MIT** **Computational Associate II**

(July 2025 – Present)

- Building novel batching strategies for single cell foundation model training.
- Using multimodal machine learning with scRNA-seq and spatial transcriptomics data to define and engineer cell states in cancer.

### Biomedical Machine Learning – Project Ex Vivo | **MICROSOFT RESEARCH** **Predoctoral Research Assistant**

(Feb 2024 – July 2025)

- Manipulating single cell foundation models to increase data depth and biology in capturing cell-state level biology
- Creating an end-to-end pipeline for alignment and analysis of single-cell RNA seq data

### Levin Lab (Genomics) | **BROAD INSTITUTE OF HARVARD/MIT** **Computational Associate Intern**

(May 2023 – September 2023)

- Integrated long-read MAS-seq data, Visium spatial transcriptomics data, and single-nucleus RNA-seq to investigate spatially organized RNA isoform diversity in a pilot study on Parkinson's disease.
- Analyzed single-nucleus RNA-seq data of a SCN2A knock-out mice cortex from the SCHEMA Schizophrenia initiative.

### Knott Lab (Genomics + ML) | **CEDARS SINAI** **Computational Associate Intern**

(May 2022 – October 2022)

- Analyzed temporal and spatial differences of cell types with scRNA-seq and ATAC-seq data from patients of an ER+ Breast Cancer clinical trial to characterize the tumor microenvironment in three different time periods to elucidate the molecular mechanisms that drive ER+ breast cancer.

### Gibson Lab | **GEORGIA TECH** **Computational Research Assistant**

(May 2021 – December 2023)

- Curated 133,488 B cells characterized from 246 tumor samples across 9 human cancer types to identify distinct molecular characteristics in the tumor microenvironment across cancer types - creation of pan-cancer atlas of B cells.
- Compared the Immune transcriptional spectrum across healthy colon, IBD, Crohn's Disease, and colon cancer progression.

### Laser Dynamics Lab | **GEORGIA TECH** **Research Assistant**

(May 2021 – May 2022)

- Computationally modeled and synthesized gold and silver plasmonic nanoparticles of different morphologies for their applications in bio-sensing and photothermal therapy in cancer treatment

### Kolliputi Lab (Immunology) | **UNIVERSITY OF SOUTH FLORIDA** **Research Writer**

(May 2020 – March 2021)

- Compared respiratory viruses MERS-CoV, SARS-CoV-1, HPIV3, RSV, IAV to SARS-CoV-2 to uncover monocyte cytokine production as a driving feature of COVID-19 infection.
- Invited to write the chapter on 'The Epigenetics of Pulmonary Disease' for the textbook Medical Epigenetics, 2E (Elsevier)

## POSTERS First Author †

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1. **Akshaya Thoutam†**, Madeline Hughes (2025, April). Evaluating the role of pre-training dataset size, diversity, and composition on single-cell foundation model performance. [RECOMB 2025 Research in Computational Molecular Biology](#).
2. **Akshaya Thoutam†**, William Watson, Mohan William, Regina Nwosu (April 2023). 5HT2c against lorcaserin on spontaneous nerve root 3 activity in crayfish model. [Georgia Tech Undergraduate Neuroscience Symposium](#)
3. Hira Anis, **Akshaya Thoutam**, Makeda Hailu, Vishal Dhere, Greg Gibson, Erin Connolly (2023, April) Single-cell Pan-cancer atlas reveals diversity and plasticity of tumor-infiltrating plasma cells. [Atlanta Workshop for Single Cell Omics 2023](#)
4. **Akshaya Thoutam†** C Greg Gibson (2022, October). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating B cells. [National Symposium for Undergraduate Research](#)
5. **Akshaya Thoutam†**, Erin Connolly, Greg Gibson (2022, April). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating Plasma Cells. [Atlanta Workshop for Single Cell Omics 2022](#).
6. **Akshaya Thoutam†**, Erin Connolly, Greg Gibson (2022, April). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating Plasma Cells. [Atlanta Workshop for Single Cell Omics 2022](#).

## TALKS

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1. Ex Vivo Joint Lab Meeting, **Microsoft Research** — State-Associated Drug Screening: Systematically Defining Dependencies Associated with Cell State Signatures. December 2025.
2. **GT Bioinformatics** and Quantitative Biosciences Panel — Career Mentoring: Industry Research and Research Internships. October 2023.
3. **Akshaya Thoutam** & Greg Gibson (2023, April). Pan-cancer atlas of tumor-infiltrating B cells. Oral presentation. Georgia Institute of Technology [Undergraduate Research Symposium](#)

## AWARDS/HONORS

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**BroadHacks 2025 Broadie's Choice Award Winners** | [Broad Institute](#)  
**College of Sciences Deans Scholar** | [Georgia Institute of Technology](#)  
**Early Research Award** | [Georgia Institute of Technology](#)  
**Robert Gunn Student Award (Finalist)** | [American Physiology Society](#)

June 2025  
Awarded every semester  
August 2021  
April 2023

## LEADERSHIP

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**Models, Inferences, Algorithms (steering committee)** | [Broad Institute](#)  
**EWSC Spatial transcriptomics study group (Organizer)** | [Broad Institute](#)  
**Down to Dance (Founder/President)** | [Georgia Institute of Technology](#)  
**Urban Health and Wellness committee leader** | [GT AMSA](#)  
**Fundraising committee leader** | [GT AMSA](#)

July 2025 – Present  
July 2026 – Present  
August 2020 – May 2023  
September 2020 – May 2021  
September 2020 – May 2021

SHADOWING

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**Pediatric Pathology** | Boston Children’s Hospital | **60 hours**  
**Pediatric Heme Oncology** | Boston Children’s Hospital | **8 hours**  
**Neurosurgery** | Atlanta Veteran Affairs Medical center | **80 hours**  
**Pediatrician** | All Pediatrics Tampa | **100 hours**  
**Sarcoma Oncology** | Moffitt Cancer Center | **40 hours**  
**Thoracic surgery** | Moffitt Cancer Center | **40 hours**

VOLUNTEERING

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|----------------------------------------------------------------------------------|-----------------------------|
| <b>Discovery center tour guide</b>   Broad Institute   <b>50 hours</b>           | July 2025 - Present         |
| <b>Teaching</b>   Harvard Medical school   <b>50 hours</b>                       | July 2025 – May 2026        |
| <b>Emergency Department</b>   Massachusetts General Hospital   <b>74 hours</b>   | August 2024 – December 2024 |
| <b>Arts in medicine Program</b>   Emory Winship Cancer Center   <b>150 hours</b> | August 2020 – May 2021      |
| <b>Care package</b>   Advent Health Florida   <b>100 hours</b>                   | March 2020 – December 2020  |
| <b>Recycling initiative</b>   Cedars Sinai Hospital   <b>200 hours</b>           | May 2022 – August 2022      |